Noisy Quantum Teleportation Channel Classification Using Machine Learning

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Abstract. Quantum teleportation (QT) is the most essential component of quantum communication, but its performance falls significantly when noisy intermediate-scale quantum (NISQ) channels are present. This paper presents a framework utilizing machine learning (ML) techniques for the classification of quantum noise channels, aimed at facilitating the optimization of real-time teleportation protocols. We employ Kraus operators to model five representative noise processes: depolarizing, amplitude damping, phase damping, bit flip, and phase flip. Additionally, we simulate their effects on teleportation fidelity and entanglement. A dataset comprising 15,360 instances is generated using the Monte Carlo method, featuring 35-dimensional vectors derived from quantum information metrics, including fidelity, entropy, purity, and Quantum Fisher Information (QFI). Multiple classifiers are evaluated, including Support Vector Machines (SVM), Random Forest (RF), Multilayer Perceptron (MLP), and Gradient Boosting (GB). The results indicate that ensemble methods demonstrate superior performance compared to alternative models. Specifically, GB attained the highest accuracy of 96.5%, ensuring strong per-class performance across all categories of errors. The results indicate that physics-informed ML is a viable approach for achieving scalable and noise-resilient QT.

Keywords: Quantum teleportation · Noise · Machine learning · Classification.

1 Introduction

Quantum teleportation is a fundamental mechanism in quantum communication that allows quantum states to be transferred between distant nodes using entanglement and classical communication. The QT, first introduced by [2], uses a shared entangled pair and classical communication to transfer an unknown quantum state (e.g., Bell state $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$). However, in ideal circumstances, it provides complete fidelity, but its actual application in quantum networks is hindered by NISQ devices that exhibit decoherence, gate faults, and channel noise [11]. Where photon loss in free space or fiber-optic channels restricts long-distance teleportation, decoherence, which is brought on by environmental interactions, weakens entanglement. For instance, studies conducted beyond 100 km have observed fidelity losses due to air turbulence and absorption [10]. Those challenges degrade entanglement, decreasing teleportation fidelity and cutting off scalability. Recent research suggests new methods for noise-resistant teleportation that provide optimal fidelity without the need for more qubits by utilizing generalized quantum measurements and control units (such as rotation gates and CNOT). But these approaches aren't scalable for large networks and have limited resources. Quantum network scalability is significantly hampered by noisy quantum channels, which are defined by amplitude damping, phase damping, or depolarizing noise. The no-cloning theorem inhibits direct signal amplification, requiring alternate methods such as quantum repeaters, which employ entanglement switching to enhance range [7].

Presently, artificial intelligence (AI) has become a significant factor in the progression of quantum communication. AI-driven techniques, specifically ML-based models including Neural Networks (NN), Reinforcement Learning (RL), and SVM, are being employed to tackle intricate quantum challenges. ML is utilized in various aspects of quantum communication, such as error correction, noise reduction, optimization of quantum channels, and distribution of entanglement [12]. Furthermore, AI facilitates the acceleration of quantum key distribution (QKD), improves protocol security, and optimizes resource management within extensive quantum networks. One of the major challenges in quantum communication is its susceptibility to environmental disturbances, including photon loss, decoherence, and quantum noise, which compromise signal integrity [3]. AI algorithms are essential for the detection, prediction, and mitigation of errors in real time, thereby boosting the reliability and efficiency of quantum communication networks. Additionally, AI enhances adaptive quantum repeater networks, thereby increasing the range of QKD for secure communication across massive distances. As quantum technologies grow, ML will be a crucial tool for creating fault-tolerant quantum networks, refining quantum protocols, and guaranteeing safe, effective quantum cryptography systems.

This research presents a framework for noisy quantum channel classification in teleportation networks. The major contribution of this research can be summarized as follows:

- The paper introduces an ML-based framework that optimizes real-time quantum teleportation protocols in the presence of noisy channels.
- We modeled five types of quantum noise utilizing Kraus operators, providing a precise simulation of their effect on teleportation fidelity.
- To improve the scalability and noise tolerance of quantum communication networks, a 35-dimensional feature vector based on quantum information was developed for the classification of noisy quantum channels.

The remaining portions of the document are arranged as follows: In Section 2, we survey relevant literature on noisy QT channel classification, and in Section

3, we lay out the methods in detail. In Section 4, we elucidate the experimental outcomes. The work is concluded in Section 5, which details the limitations and offers suggestions for future research.

2 Realted Work

In today's world, quantum systems can be precisely controlled to carry out new kinds of operations by taking advantage of their coherence features, like entanglement and superposition. But ambient noise causes quantum operations to lose precision and decoherence, hence quantum behaviour disappears. The paper [5] exhibited a Gaussian Mixture Model-Expectation Maximization (GMM-EM) method for modeling hybrid quantum noise (Poisson shot noise + AWGN) in Gaussian quantum channels. The approach captures overlapping, variable-weighted noise distributions more accurately than K-means or DBSCAN by lowering and optimizing Gaussian components. Simulations on synthetic bivariate HQN data reveal that GMM-EM produces improved clustering metrics (AIC, BIC, silhouette) and the greatest possible rates for both general and satellite quantum channels, with significant increases at 25 dB SNR. Another study [13] examined Bell, GHZ, and W states in the context of phase damping, amplitude damping, and depolarizing noise. It demonstrates that Bell states exhibit the highest fidelity in certain damping scenarios, whereas GHZ and W states present greater resilience in the presence of multi-qubit noise. The findings provide insights for the selection of optimal channels in noisy quantum communication. During this study [8], the authors developed a hybrid classification model that integrates classical preprocessing with quantum circuits, addressing amplitude and phase damping noise as represented by Kraus operators. In the Iris dataset, features are encoded into quantum states, processed via dissipative two-qubit channels, then measured for classification purposes. The model attains an accuracy of up to 96% in low-noise environments, demonstrating enhanced resilience to amplitude damping and significant promise for NISQ applications. One more study [1] showed a hybrid model that integrates BB84 and BBM92 QKD methods with classical ML to detect amplitude damping, bit-flip, and depolarizing noise utilizing QBER-based features and principal component analysis. The K-Nearest Neighbors (KNN), Gaussian Naive Bayes (GNB), and SVM were evaluated, attaining up to 96% accuracy in remote-channel scenarios and exceeding 99% in gate-based simulations, with SVM hitting 100%, illustrating that basic QKD with ML can effectively discern quantum noise.

Additionally, this paper [6] investigated a noise-aware Variational Quantum Algorithm framework that trains parameterized circuits for variational ground-state preparation under depolarizing, amplitude, and phase damping noise. It uses circuit compilation to create device-specific hardware-efficient implementations. The approach delivers up to 96% accuracy on noisy simulators and increased hardware fidelity on classification and state preparation tasks, demonstrating the benefits of noise modeling and compilation optimization. Here, the author [14] specified an effective technique for studying the model and detecting the noise of quantum channels using a quantum neural network (QNN). They develop the architecture of the QNN and the model training procedure based on different kinds of noisy quantum channel models. The cost function of the QNN may approach 90% of the channel model, according to the findings of our experiments, which we conducted to confirm the scheme's training efficacy. A different study [4] investigated a QT error mitigation strategy utilizing quantum topological semion codes and a convolutional neural network (CNN) decoder to safeguard states against noise. The double-semion code is represented in a toric-code-like 8×8 configuration, enabling the CNN to forecast logical-level corrections inside a depolarizing noise framework. It is trained on diverse error rates, attaining thresholds of 0.082 (< 0.007 dB) and 0.096 (< 0.01 dB), exceeding most classical decoders. Subsequent research stipulated a technique for encoding qubits in two time intervals utilizing "time-bin entanglement" for long-distance quantum communication. This method protects qubits from noise without needing more resources as the channel length grows, which keeps the fidelity high. The setup, which includes optical parts like half-wave plates and polarizing beam splitters, can be tested and allows for safe, high-quality transmission [9].

3 Methodology

This section presents the specific steps to take in order to classify quantum channels in noisy QT networks using a combination of ML and quantum information theory. By integrating theoretical and experimental factors, the method aims to facilitate the optimization of protocols in real-time. Modeling quantum channels, simulating the QT protocol, extracting features using quantum information metrics, and classifying quantum channels using ML are the components that make up the methodology.

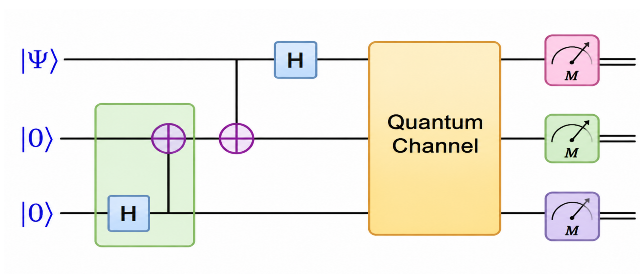

Fig. 1. Proposed workflow for noisy quantum teleportation channel classification.

The circuit in Figure 1 illustrates an approach for quantum communication, most likely for QT. After the qubits in the $|0\rangle$ state are prepared, they are put through Hadamard and CNOT gates to generate entanglement between them. The entangled qubits are transmitted across the quantum channel, and the qubits undergo final measurements, denoted by M. Measurements bring the quantum state to a collapse, which enables classical communication to finish teleporting the state $|\psi\rangle$. Here we see the fundamental quantum communication phases of entanglement distribution, quantum state transmission, and measurement.

3.1 Quantum Channel Modeling and Noise Analysis

Modeling quantum channels is the first and most important part of our methodology. This is necessary for simulating realistic noise circumstances in QT networks. Here, quantum channels are represented by Kraus operators, which stand for the development of quantum states subjected to different types of noise variations. We simulate several kinds of noise, each of which stands for a distinct physical mechanism that degrades the quantum state throughout transmission.

Pauli Matrices and Quantum States A single qubit's quantum state is denoted by a density matrix ρ, defined as:

$$\rho = |\psi\rangle\langle\psi| \tag{1}$$

where $|\psi\rangle$ is a quantum state vector that has been normalized. When different Pauli matrices are used, the development of the quantum state changes. The bit-flip, phase-flip, and depolarizing noise are represented by the Pauli matrices X, Y, Z:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2}$$

These matrices are essential for the modeling of noise and quantum error correction, as they characterize the primary quantum errors that manifest in communication channels.

Kraus Operators for Quantum Channel Types We define various quantum channels that represent typical QT noise sources. To characterize the quantum state evolution as a result of these noise processes, the Kraus operators for each channel are constructed. Here are some quantum channels that are being considered:

Depolarizing Channel This channel introduces stochastic errors and is modeled as follows:

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}\left(X\rho X^\dagger + Y\rho Y^\dagger + Z\rho Z^\dagger\right) \tag{3}$$

where p is the probability of error, and X, Y, Z are the Pauli matrices.

Amplitude Damping Channel This channel simulates energy loss (T1 decay) and is characterized by the subsequent Kraus operators:

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} \tag{4}$$

Here, the damping parameter γ measures the amount of energy lost to the environment.

Phase Damping Channel This channel characterizes the T2* decay, a loss of quantum coherence, and its Kraus operators are:

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\gamma} \end{pmatrix} \tag{5}$$

The parameter γ once more regulates the rate of coherence loss.

Bit Flip and Phase Flip Channels For modeling these channels, the Pauli matrices X (bit flip) and Z (phase flip) are used on the density matrix ρ. The generic form is:

$$\mathcal{E}_X(\rho) = (1-p)\rho + p(X\rho X^\dagger) \tag{6}$$

$$\mathcal{E}_Z(\rho) = (1-p)\rho + p(Z\rho Z^\dagger) \tag{7}$$

where p is the error probability.

These noise models represent the degradation of quantum states during teleportation, with the parameters p or γ regulating the intensity of the noise.

3.2 Quantum Teleportation Protocol Simulation

Quantum teleportation is a procedure in which the quantum state of a qubit is transmitted from one party to another through the utilization of entanglement and classical communication. Our simulation adheres to the following general steps:

State Preparation A single qubit is developed in a superposition state $|\psi\rangle$, defined by two angles on the Bloch sphere, θ and ϕ. The quantum state can be represented as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \tag{8}$$

Bell Pair Generation A maximally entangled *Bell state* is produced between two parties, Alice and Bob. This entanglement serves as a resource throughout the teleportation process.

Channel Transmission The quantum state $|\psi\rangle$ is transmitted through a noisy quantum channel. We use the Kraus operators that were previously presented to model the noise.

Bell Measurement and Classical Communication Alice uses her qubit to do a Bell-state measurement and then transmits the result to Bob over a classical communication channel. Bob uses a unitary operation on his qubit, which is determined by the measurement result.

State Reconstruction Bob performs an appropriate unitary transformation on his qubit, effectively restoring the teleported state $|\psi\rangle$.

Fidelity Calculation The fidelity F of the teleported state in comparison to the original state is determined using the formula:

$$F(\rho, \rho') = \mathrm{Tr}\left(\sqrt{\sqrt{\rho}\rho'\sqrt{\rho}}\right)^2 \tag{9}$$

Where ρ represents the starting point and ρ' denotes the final point after teleportation.

3.3 Quantum Information Metrics

To assess the quality of the teleportation process under varying noise conditions, we calculate several essential quantum information metrics that offer insights into the system's state.

Fidelity The fidelity of a quantum system is defined as its proximity to the ideal state in its ultimate state. This metric is used to evaluate the effectiveness of teleportation and is described as:

$$F(\rho, \rho') = \mathrm{Tr}\left(\sqrt{\sqrt{\rho}\rho'\sqrt{\rho}}\right)^2 \tag{10}$$

Trace Distance The trace distance among two quantum states, represented as ρ and ρ', is computed using the subsequent formula:

$$T(\rho, \rho') = \frac{1}{2}\mathrm{Tr}|\rho - \rho'| \tag{11}$$

This measurement shows how distinguished the two quantum states are from one another.

Von Neumann Entropy The Von Neumann entropy $S(\rho)$ quantifies the mixedness of a quantum state:

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho) \tag{12}$$

This is particularly useful when trying to estimate how much data gets lost while teleporting.

Purity The purity $\gamma(\rho)$ is defined as:

$$\gamma(\rho) = \mathrm{Tr}(\rho^2) \tag{13}$$

Purity quantifies the degree to which a quantum state is pure, indicating its freedom from noise.

Quantum Fisher Information (QFI) It quantifies how much a quantum state changes in response to a parameter change. The expression is provided by:

$$\mathcal{F}(\rho, \theta) = \sum_{i,j} \frac{2|\langle\psi_i|\frac{\partial\rho}{\partial\theta}|\psi_j\rangle|^2}{\lambda_i + \lambda_j} \tag{14}$$

where λ_i are the eigenvalues of the state ρ, and $\frac{\partial\rho}{\partial\theta}$ is the derivative of the density matrix with respect to the parameter θ.

3.4 Dataset Overview

We developed a comprehensive dataset by conducting extensive Monte Carlo simulations of QT methods under various noise models. The dataset comprises 15,360 unique instances of quantum channels, with each instance categorized into one of five distinct channel types: depolarizing, amplitude damping, phase damping, bit flip, and phase flip. Each sample is characterized by a 35-dimensional feature vector, which includes physics-informed descriptors obtained from quantum information theory, such as fidelity measures, entanglement metrics, and parameters that can be experimentally adjusted. To achieve a balanced classification task, it is essential that each channel type is represented equally, with a total of 3,072 instances assigned to each type. The parameter space was methodically sampled to encompass a broad spectrum of noise intensities and initial states throughout the Bloch sphere, thus ensuring a thorough representation of realistic variations in quantum systems.

3.5 Machine Learning for Quantum Channel Classification

We employ ML methods to categorize quantum channels using features retrieved from the teleportation protocol model. These features encompass physical factors such as θ, ϕ, and noise strength, as well as quantum information metrics such as fidelity, trace distance, entropy, and coherence loss.

Feature Selection and Engineering Feature selection based on mutual information and analysis of variance (ANOVA) allows us to identify the most distinctive features. These features that were chosen are then standardized using StandardScaler to make sure they are uniform and to facilitate ML simpler.

Machine Learning Models We employ the following ML models for the classification of quantum channels:

Random Forest (RF) The RF is an ML technique that uses several decision trees (DT) to reduce correlation between feature data. Concurrently, the computational cost of RF is O(n) (where n denotes the number of samples) when processing vast quantities of data. This integration also allows for the approach to be conducted in parallel, which increases speed. It uses a random selection of features and samples to reduce the correlation across DT. At first, a random selection of data of a similar quantity is made from the training sample within the first training set. In addition, the decision tree is built using a randomly selected subset of the features. By reducing the correlation among each DT, these two types of randomization improve the model's accuracy and reduce the likelihood of overfitting errors.

Support Vector Machine (SVM) The SVM classifier operates by identifying an ideal separation border, known as a hyperplane, between distinct classes within the dataset. It concentrates on the nearest data points adjacent to the boundary, referred to as support vectors, and endeavours to optimize the margin between the classes. An increased margin enhances the model's accuracy in classifying new data. When the data cannot be separated linearly, SVMs employ kernel functions to raise the data's dimensionality and make separation easier. After training, the model uses the sample's location on the hyperplane to predict the class of a new sample.

Gradient Boosting (GB) The GB classifier employs a sequential ensemble learning approach, wherein numerous weak base learners, often shallow DT, are amalgamated to create a robust predictive model. The algorithm constructs an additive model incrementally by fitting each novice learner to the residual errors or negative gradients of the loss function produced by the prior ensemble. In each iteration, the model minimizes a differentiable objective function, such as logistic loss for classification, by gradient descent in function space. Misclassified or inaccurately forecasted data are prioritized in succeeding rounds, enabling the ensemble to enhance its decision limits incrementally. The final classification is achieved by adding up the weighted contributions of each tree, which is usually followed by a transformation of probabilities and class assignment. The GB attains elevated predictive accuracy and strong performance on intricate, non-linear datasets using an iterative error-correction technique.

Multilayer Perceptron (MLP) The MLP classifier was utilized as a feed-forward neural network to discern nonlinear associations between input data and target classes. Each neuron in the fully connected hidden layers processed the normalized input variables by applying a nonlinear activation function and computing a weighted sum with bias. The model parameters were refined by minimizing the classification loss, including binary or categorical cross-entropy. During training, gradients were calculated by backpropagation and weights were updated using a gradient-based optimizer. Through repeated learning over several epochs, the output layer used sigmoid or softmax activation to create class probabilities. The final classification was determined by the probability that was most accurately predicted.

4 Result Analysis

The following tables provide a detailed performance evaluation of our applied models. The overall performance of SVM, MLP, RF, and GB classifiers is summarized in Table 1. To ensure the robustness of the ML models, we perform statistical analysis of the cross-validation scores using confidence intervals and t-tests. This allows us to assess the reliability of the models and make confident comparisons between them. Also, the per-class precision, recall, and F1-scores of the best-performing GB model are highlighted in Table 2.

4.1 Evaluation Metrics

We used the following evaluation metrics to measure the proposed categorization models' performance and make sure it was a thorough and accurate assessment of their predicted efficiency.

Accuracy Accuracy is defined as the ratio of accurately predicted instances to the total expected instances. The prediction must encompass not only True Positives (TP) and True Negatives (TN) but also False Positives (FP) and False Negatives (FN). Thus, accuracy may provide an excellent approximation of the ML model's capacity to forecast unbalanced data, but precision and recall are also need to validate the accuracy metric's findings. The accuracy is delineated as per equation:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

Precision Precision is the proportion of accurately anticipated positive instances to the total expected positive instances. It is associated with low False Positive Rate (FPR) to have high precision. The precision is measured using the subsequent equation.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

Recall Recall is a metric utilised to assess the efficacy of a classification approach. It is particularly crucial when the cost of FN is substantial, indicating that overlooking a positive case is more harmful than erroneously labelling a negative case.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

F1 score The F1 score is calculated by averaging the precision and recall scores. To evaluate the classifier's efficacy, it is employed as a statistical measurement. This score considers both the positive and negative outcomes. The formula below defines the F1 score.

$$\text{F1 score} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{18}$$

Table 1. Model performance summary with Cross-Validation (CV) Mean, Confidence Interval (CI), F1-Score, and Test accuracy.

Model	**CV Mean ± Std**	**95% CI**	**F1-Score**	**Test Acc.**
SVM(RBF Kernel)	0.8900 ± 0.0118	[0.8811, 0.8988]	0.8912	0.8789
Multilayer Perceptron	0.9257 ± 0.0082	[0.9195, 0.9318]	0.9258	0.9271
Random Forest	0.9559 ± 0.0064	[0.9510, 0.9607]	0.9558	0.9577
Gradient Boosting	**0.9615 ± 0.0062**	**[0.9568, 0.9662]**	**0.9615**	**0.9648**

Table 1 compares the performance of four ML models according to cross-validation and independent test findings. GB had the best overall performance, with the highest cross-validation mean accuracy (0.9615 ± 0.0062), narrowest confidence interval [0.9568, 0.9662], test accuracy of 0.9648, and F1-score of 0.9615, indicating strong generalization ability. Furthermore, RF followed closely with a cross-validation mean of (0.9559 ± 0.0064), confidence interval [0.9510, 0.9607], test accuracy of 0.9577, and F1-score of 0.9558, indicating stable and competitive predictive power. On the other hand, the MLP model fared moderately, with a CV mean of (0.9257 ± 0.0082), confidence interval [0.9195, 0.9318], test accuracy of 0.9271, and F1-score of 0.9258, whereas the SVM (RBF kernel) did poorly, with a CV mean of (0.8900 ± 0.0118), wider confidence intervals of [0.8811, 0.8988], and the lowest test accuracy of 0.8789 and F1-score of 0.8912. Overall, ensemble-based algorithms (GB and RF) beat MLP and SVM, demonstrating their reliability and effectiveness for this task.

Table 2 delineates the class-specific performance metrics of the GB classifier, including precision, recall, and F1-score. The model attained optimal performance for the depolarizing class, exhibiting flawless precision, recall, and F1-

Table 2. Per-Class Precision, Recall, and F1-Score for the GB classifier.

Class	Precision	Recall	F1-Score
Depolarizing	1.00	1.00	1.00
Bit Flip	0.99	1.00	1.00
Amplitude Damping	0.96	0.92	0.94
Phase Damping	0.94	0.95	0.95
Phase Flip	0.93	0.95	0.94

score values of 1.00, signifying error-free classification for this category. The bit flip class had nearly flawless performance, achieving a precision of 0.99 and both recall and F1-score values of 1.00, indicating an exceedingly low incidence of false-positive predictions.

Among the remaining classes, phase damping attained an F1-score of 0.95, somewhat surpassing both amplitude damping and phase flip, which obtained F1-scores of 0.94. The amplitude damping class exhibited a precision of 0.96, although a relatively lower recall of 0.92. The phase flip class exhibited the lowest precision value of 0.93, while its recall remained elevated at 0.95. The results provide robust and equitable classification performance, with only slight deterioration for amplitude damping and phase flip, in contrast to the practically flawless depolarizing and bit flip categories.

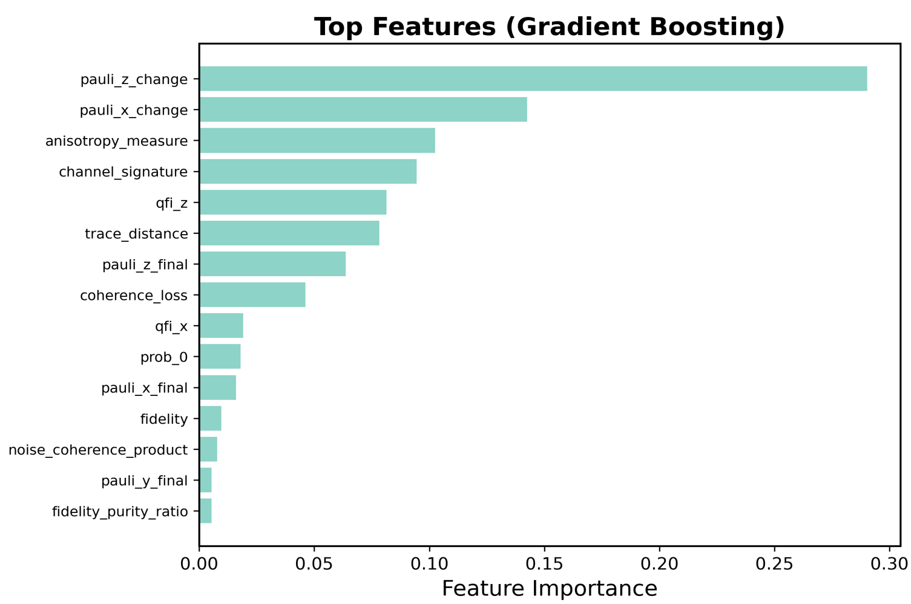

Fig. 2. Most important categorization features that are assessed via GB.

Figure 2 illustrates the most significant features identified by the GB model, ranked according to their importance in predicting the quantum noise classification task. Furthermore, as shown in Figure 3, the training and cross-validation scores improve with increasing training set size, suggesting that the model generalizes effectively.

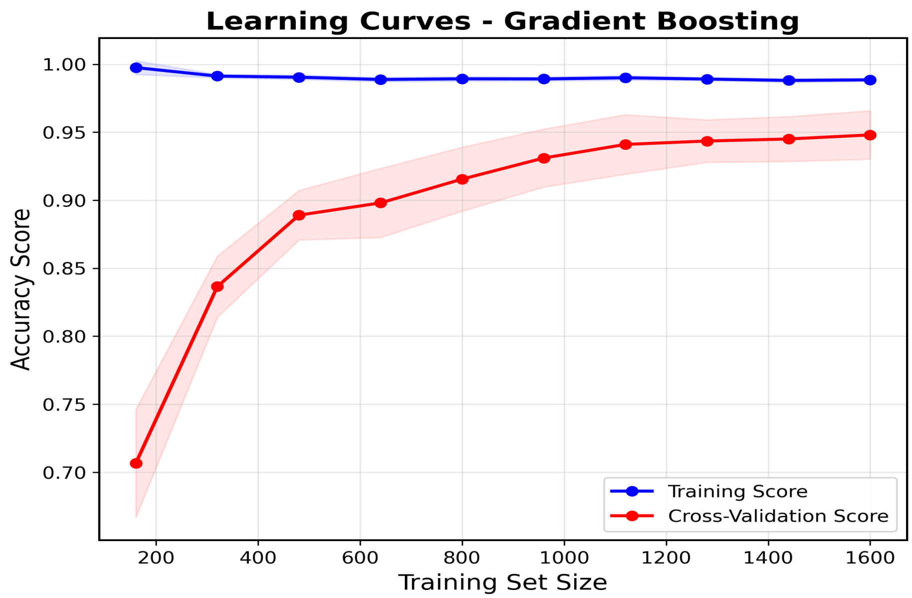

Fig. 3. Learning performance of the GB classifier, comparing training accuracy with cross-validation accuracy.

Additionally, it can be observed from Figure 4 that the average fidelity decreases for all types of noise as the noise strength increases. Although bit flip and phase flip exhibit the most significant decreases, depolarizing and amplitude-damping channels exhibit the highest levels of reliability.

5 Conclusion and Future Work

This paper presents an ML-based framework designed for the classification of noisy quantum channels within teleportation networks that facilitate the real-time optimization of quantum protocols. The framework models five different types of quantum noise by employing Kraus operators. Using this modeling approach, we simulate QT under various noise conditions, generating a comprehensive dataset consisting of 15,360 instances. Each data instance is represented by a 35-dimensional feature vector derived from key quantum information metrics, including fidelity, entropy, purity, and QFI, ensuring that the feature set

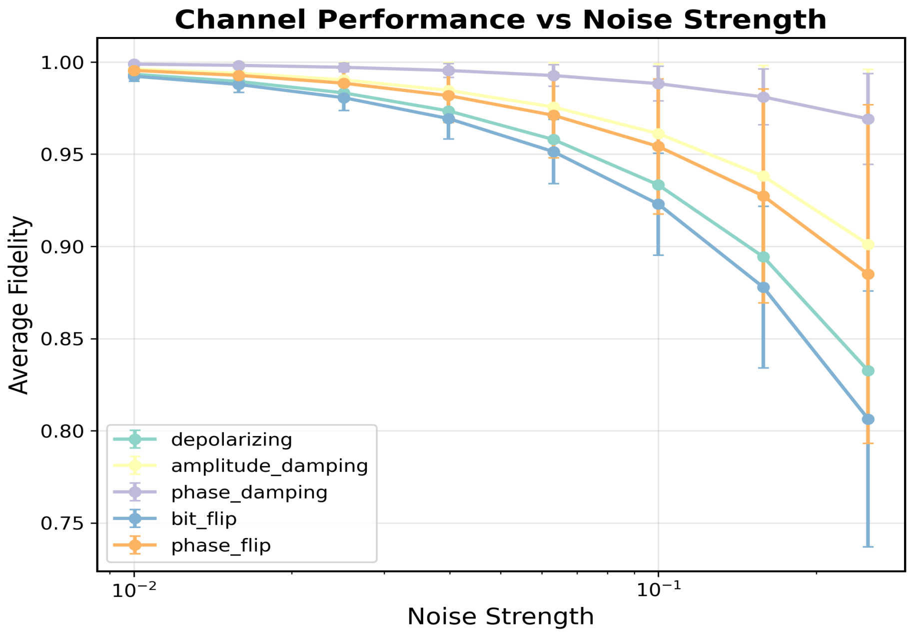

Fig. 4. Fidelity-based comparison of quantum noise channels under varying noise levels.

is physically informed and captures the underlying behavior of quantum channels. A thorough evaluation of multiple ML classifiers reveals that the ensemble methods, particularly GB and RF, outperform traditional models such as MLP and SVM. Specifically, GB demonstrates superior performance, achieving a test accuracy of 96.48% and an average F1-score of 96.6%, highlighting its effectiveness in classifying quantum channels under noisy conditions. These results substantiate the use of physics-informed ML as an effective tool for enhancing the scalability and noise resilience of QT, thus making it viable for deployment in larger, more complex quantum communication networks.

Looking ahead, future work will focus on the integration of hybrid quantum-classical models, with the aim of further improving the performance of the framework. Additionally, we plan to validate the methodology on experimental real quantum hardware, thereby addressing the challenges of real-world implementation. These advances will be crucial for enabling real-time optimization of teleportation protocols, which is a key requirement for scalable and robust quantum communication systems.

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